

Mathematical Properties of Affine-DDPNM: Corrected Statements and Proofs

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The results below are stated at the finite-dimensional discrete level. The local-kernel and global-SPD conclusions are conditional on an explicit rank condition. Galerkin optimality is proved relative to a separately defined full traction-trace system and only in its energy seminorm. Equivalence with the monolithic continuous-pressure Taylor–Hood method is not asserted.

1 Discrete setting and sign convention

For each pore Ω_i , let $A_i \in \mathbb{R}^{n_{u,i} \times n_{u,i}}$ and $B_i \in \mathbb{R}^{n_{p,i} \times n_{u,i}}$ be the matrices associated with

$$a_i(\mathbf{u}, \mathbf{v}) = \nu \int_{\Omega_i} \nabla \mathbf{u} : \nabla \mathbf{v} \, dx, \quad b_i(\mathbf{v}, q) = - \int_{\Omega_i} q \nabla \cdot \mathbf{v} \, dx.$$

Thus a local load f is solved through

$$K_i \begin{bmatrix} U \\ P \end{bmatrix} := \begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U \\ P \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}. \quad (1)$$

Each planar internal interface γ_k has a global frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$. For the two adjacent pores, use oriented directions $\mathbf{d}_{i,k,\beta}$ satisfying

$$\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}, \quad \beta \in \{n, t_1, t_2\}. \quad (2)$$

One code-compatible choice is

$$\mathbf{d}_{i,k,n} = \mathbf{n}_{i,k}, \quad \mathbf{d}_{i,k,t_\nu} = -\sigma_{i,k} \mathbf{t}_{k,\nu}.$$

Let $\varphi_0 = 1$, $\varphi_s = s_k$, and $\varphi_t = t_k$, with $\mathcal{M} = \{0, s, t\} \times \{n, t_1, t_2\}$.

The coefficients parameterize inward traction:

$$-\boldsymbol{\sigma}(\mathbf{u}_i, p_i) \mathbf{n}_{i,k} = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{k,\alpha,\beta} \varphi_\alpha \mathbf{d}_{i,k,\beta}. \quad (3)$$

Since the weak form uses the physical outward traction $\boldsymbol{\sigma} \mathbf{n}_{i,k}$, the primitive load column is

$$[f_{i,k}^{(\alpha,\beta)}]_j = - \int_{\gamma_k} \varphi_\alpha \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \, dS. \quad (4)$$

Let $L_i \in \mathbb{R}^{n_{u,i} \times n_i}$ collect all local primitive columns, including one constant-normal column for each known boundary port.

Assumption 1.1 (Local algebraic stability). *For every pore i :*

(a) A_i is symmetric positive definite;

- (b) B_i has full row rank;
- (c) the coefficient vector $\mathbf{1}_i^p$ of the constant pressure $q_h \equiv 1$ belongs to the chosen local pressure space.

The first item is normally obtained from a nontrivial Dirichlet wall segment. The second item is an independent discrete inf-sup/rank requirement and must not be inferred from the first item alone.

Define

$$M_i = B_i A_i^{-1} B_i^\top, \quad H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}. \quad (5)$$

Under Assumption 1.1, M_i is SPD.

2 Local response operator

Lemma 2.1 (Local well-posedness). *Under Assumption 1.1, the saddle matrix K_i in (1) is invertible. For every f , the velocity component is $U = H_i f$ and satisfies $B_i U = 0$.*

Proof. Since A_i is SPD and B_i has full row rank, the pressure Schur complement $M_i = B_i A_i^{-1} B_i^\top$ is SPD. Standard block elimination gives

$$P = M_i^{-1} B_i A_i^{-1} f, \quad U = A_i^{-1} (f - B_i^\top P) = H_i f.$$

Hence the solution is unique. Direct substitution gives $B_i H_i = 0$. $\square$

Lemma 2.2 (Structure of H_i). *The matrix H_i is symmetric positive semidefinite and satisfies*

$$H_i^\top A_i H_i = H_i, \quad (6)$$

$$\text{Ker}(H_i) = \text{Range}(B_i^\top). \quad (7)$$

Moreover, for every f ,

$$f^\top H_i f = (H_i f)^\top A_i (H_i f) \geq 0. \quad (8)$$

Proof. Symmetry follows from (5). Expanding $H_i A_i H_i$ and using $M_i = B_i A_i^{-1} B_i^\top$ gives (6); hence (8) and positive semidefiniteness follow.

Also $H_i B_i^\top = 0$, so $\text{Range}(B_i^\top) \subseteq \text{Ker}(H_i)$. Conversely, if $H_i f = 0$, Lemma 2.1 gives velocity $U = 0$ in (1). The first block row then yields $f = B_i^\top P$ for the corresponding pressure vector P . Therefore $\text{Ker}(H_i) \subseteq \text{Range}(B_i^\top)$. $\square$

Define the local compliance matrix

$$\mathbf{G}_i = L_i^\top H_i L_i. \quad (9)$$

For a coefficient vector λ_i , the reconstructed velocity is $U_i = H_i L_i \lambda_i$.

Lemma 2.3 (Symmetry, energy identity, and exact kernel formula). *The matrix $\mathbf{G}_i$ is symmetric positive semidefinite and*

$$\lambda_i^\top \mathbf{G}_i \lambda_i = U_i^\top A_i U_i \geq 0. \quad (10)$$

Its kernel is exactly

$$\text{Ker}(\mathbf{G}_i) = \text{Ker}(H_i L_i) = \{\lambda_i : L_i \lambda_i \in \text{Range}(B_i^\top)\}. \quad (11)$$

Proof. Symmetry and (10) follow from Lemma 2.2:

$$\lambda_i^\top L_i^\top H_i L_i \lambda_i = (H_i L_i \lambda_i)^\top A_i (H_i L_i \lambda_i).$$

Therefore $\lambda_i \in \text{Ker}(\mathbf{G}_i)$ if and only if $H_i L_i \lambda_i = 0$. The second equality in (11) follows from (7). No claim that H_i is positive definite on $\text{Range}(L_i)$ is needed; in fact that claim would be false whenever the constant-pressure load belongs to $\text{Range}(L_i)$. $\square$

3 Constant-pressure mode and the conditional local-kernel theorem

Let $\mathbf{1}_{i,0n} \in \mathbb{R}^{n_i}$ have value one in every constant-normal port slot, including known boundary ports, and zero in all affine-linear or tangential slots.

Lemma 3.1 (Constant-pressure null vector). *With the load convention (4),*

$$L_i \mathbf{1}_{i,0n} = B_i^\top \mathbf{1}_i^p. \quad (12)$$

Consequently $\mathbf{1}_{i,0n} \in \text{Ker}(\mathbf{G}_i)$.

Proof. For a velocity basis function $\phi_j^{(i)}$, summing all constant-normal port columns gives

$$\begin{aligned} [L_i \mathbf{1}_{i,0n}]_j &= - \int_{\partial\Omega_i} \phi_j^{(i)} \cdot \mathbf{n}_i dS \\ &= - \int_{\Omega_i} \nabla \cdot \phi_j^{(i)} dx = [B_i^\top \mathbf{1}_i^p]_j, \end{aligned}$$

because B_i represents $-\int q \nabla \cdot \mathbf{v}$. Now apply Lemma 2.2. $\square$

Assumption 3.2 (No nonconstant pressure-load dependency). *For each pore,*

$$L_i \lambda_i \in \text{Range}(B_i^\top) \implies \lambda_i = c \mathbf{1}_{i,0n} \text{ for some } c \in \mathbb{R}. \quad (13)$$

Equivalently,

$$\text{rank}(H_i L_i) = n_i - 1. \quad (14)$$

Theorem 3.3 (Conditional local-kernel characterization). *Under Assumptions 1.1 and 3.2,*

$$\text{Ker}(\mathbf{G}_i) = \text{span}\{\mathbf{1}_{i,0n}\}. \quad (15)$$

Proof. Lemma 3.1 gives inclusion of the right-hand side in the left. Conversely, Lemma 2.3 implies that any $\lambda_i \in \text{Ker}(\mathbf{G}_i)$ satisfies $L_i \lambda_i \in \text{Range}(B_i^\top)$. Assumption 3.2 then gives $\lambda_i = c \mathbf{1}_{i,0n}$. $\square$

Remark 3.4 (What the rank assumption means). *Equation (14) says that the n_i response columns $H_i f_i^{(\mu)}$ have exactly one linear dependency. It does not say that they have codimension one in the full space $\text{Range}(H_i)$; that codimension is generally $\text{rank}(H_i) - (n_i - 1)$ and may be large. Independence of the nine functions on each individual face is not sufficient for Assumption 3.2, because cross-port dependencies and dependencies introduced by H_i remain possible.*

4 Global Schur complement, uniqueness, and mass conservation

Let $\lambda \in \mathbb{R}^{9m}$ collect the shared unknown interface blocks. For pore i , write $\lambda_i^u = R_i^u \lambda$. Let $J_i^u : \mathbb{R}^{9m_i^u} \rightarrow \mathbb{R}^{n_i}$ be zero extension into all known boundary slots:

$$J_i^u z = (z, 0).$$

Then

$$\mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u \quad (16)$$

and, for any global vector z ,

$$z^\top \mathbf{S} z = \sum_{i=1}^N (J_i^u R_i^u z)^\top \mathbf{G}_i (J_i^u R_i^u z). \quad (17)$$

Assumption 4.1 (Graph anchoring). *Every connected component of the pore adjacency graph contains at least one pore with at least one known inlet/outlet traction port.*

Theorem 4.2 (Global SPD and unique solvability). *Under Assumptions 1.1, 3.2, and 4.1, the matrix $\mathbf{S}$ is symmetric positive definite. Therefore the affine Schur system has a unique solution for every right-hand side.*

Proof. Symmetry is immediate. By (17) and Lemma 2.3, $\mathbf{S}$ is positive semidefinite.

Suppose $z^\top \mathbf{S} z = 0$. Every nonnegative summand in (17) vanishes, so Theorem 3.3 gives

$$J_i^u R_i^u z = c_i \mathbf{1}_{i,0n} \quad (18)$$

for each pore. If pore i has a known boundary port, the corresponding component on the left of (18) is zero, whereas that on the right is c_i ; hence $c_i = 0$.

If pores i and j share an internal interface, the same global constant-normal coefficient occurs in both restrictions $R_i^u z$ and $R_j^u z$. Equation (18) therefore implies $c_i = c_j$. The constants are thus equal throughout each connected graph component. By Assumption 4.1, each component contains a pore with a known port, where the constant is zero. Hence every $c_i = 0$ and all internal components of z vanish. Thus $z = 0$. $\square$

Remark 4.3 (Necessity of the hypotheses). *Without graph anchoring, a connected component supports a global uniform constant-normal coefficient and $\mathbf{S}$ is singular. Without Assumption 3.2, additional local null directions can survive and may produce global singularity. Thus the unconditional SPD statement in the original methodology is not justified by per-face independence alone.*

For a local reconstructed velocity, define

$$M_{i,k}^{(\alpha,\beta)} = \int_{\gamma_k} \varphi_\alpha \mathbf{u}_{i,h} \cdot \mathbf{d}_{i,k,\beta} dS.$$

By (4), $L_i^\top U_i$ is the negative vector of these moments. The assembled Schur equations enforce $M_{i,k}^{(\alpha,\beta)} + M_{j,k}^{(\alpha,\beta)} = 0$.

Theorem 4.4 (Discrete mass conservation). *Under Assumption 1.1, every reconstructed local response satisfies*

$$\sum_{k \subset \partial\Omega_i} M_{i,k}^{(0,n)} = 0. \quad (19)$$

If the global Schur equations are solved exactly, internal constant-normal moments cancel pairwise and

$$\int_{\partial\Omega} \mathbf{u}_h^{\text{DD}} \cdot \mathbf{n} dS = 0. \quad (20)$$

Proof. The second block row of (1) gives $B_i U_i = 0$. Testing with the constant pressure coefficient $\mathbf{1}_i^p$ yields

$$0 = (\mathbf{1}_i^p)^\top B_i U_i = - \int_{\Omega_i} \nabla \cdot \mathbf{u}_{i,h} dx = - \int_{\partial\Omega_i} \mathbf{u}_{i,h} \cdot \mathbf{n}_i dS,$$

which is (19). Summing over pores and using the enforced interface moment continuity cancels all internal fluxes, leaving only the external boundary flux. $\square$

5 Full traction-trace reference and affine Galerkin restriction

This section defines the reference problem needed for a valid enrichment statement. It is a dual traction-trace system using the same local matrices A_i, B_i and therefore the same local pressure spaces. It is not identified with the primal exact FE-trace Schur complement of the monolithic continuous-pressure problem.

On interface γ_k , let $X_{k,h}$ be a scalar trace space with basis $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$. A full vector traction coefficient is $\xi_k = (\xi_k^n; \xi_k^{t_1}; \xi_k^{t_2}) \in \mathbb{R}^{3m_k}$. Using the same oriented directions as in (2), define the local coupling blocks

$$[N_{i,k}^\beta]_{\ell j} = \int_{\gamma_k} \psi_{k,\ell} \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} dS. \quad (21)$$

The physical weak load for inward-traction coefficients is

$$\widehat{L}_i \xi_i = -N_i^\top \xi_i. \quad (22)$$

The local and assembled full-trace matrices are

$$\widehat{S}_i = N_i H_i N_i^\top, \quad \widehat{S} = \sum_i (\widehat{R}_i^u)^\top \widehat{S}_i^{uu} \widehat{R}_i^u. \quad (23)$$

They are symmetric positive semidefinite. Define $\|x\|_{\widehat{S}} = (x^\top \widehat{S} x)^{1/2}$; without an additional full-trace kernel theorem this is a seminorm.

5.1 Correct affine restriction matrix

Let M_k be the scalar face mass matrix and

$$[b_k^\alpha]_\ell = \int_{\gamma_k} \varphi_\alpha \psi_{k,\ell} dS, \quad p_k^\alpha = M_k^{-1} b_k^\alpha, \quad \alpha \in \{0, s, t\}.$$

If the affine scalar functions belong to $X_{k,h}$, this projection represents them exactly. For coefficient order

$$(0, n), (0, t_1), (0, t_2), (s, n), (s, t_1), (s, t_2), (t, n), (t, t_1), (t, t_2),$$

the correct restriction is

$$P_k^{\text{aff}} = \begin{bmatrix} p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 & 0 \\ 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 \\ 0 & 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 9}. \quad (24)$$

In particular, a tangential column has no normal block. Let $P_{\text{aff}} = \bigoplus_k P_k^{\text{aff}}$.

Lemma 5.1 (Algebraic Galerkin equivalence). *Assume the same oriented trace basis and restriction maps are used in the full and affine systems. Then*

$$\mathbf{S} = P_{\text{aff}}^\top \widehat{S} P_{\text{aff}}, \quad F = P_{\text{aff}}^\top \widehat{F}. \quad (25)$$

Proof. On each pore, (4), (21), and (24) give the matrix identity

$$L_i^u = -(N_i^u)^\top P_i^{\text{aff}}.$$

Therefore

$$\mathbf{G}_i^{uu} = (L_i^u)^\top H_i L_i^u = (P_i^{\text{aff}})^\top N_i^u H_i (N_i^u)^\top P_i^{\text{aff}}.$$

The minus signs cancel in the quadratic matrix. Signed Boolean assembly commutes with the block-diagonal restriction because both descriptions use the same globally shared traction coefficients and the same opposite local directions. Summing over pores proves the first identity. The known-port cross term gives the second identity with the same sign convention. $\square$

Theorem 5.2 (Best approximation in the full-trace energy seminorm). *Assume the full system $\widehat{S}\xi^{\text{full}} = \widehat{F}$ is consistent. Let λ^{aff} solve $\mathbf{S}\lambda^{\text{aff}} = F$ and set $\xi^{\text{aff}} = P_{\text{aff}}\lambda^{\text{aff}}$. Then*

$$\|\xi^{\text{full}} - \xi^{\text{aff}}\|_{\widehat{S}} = \min_{\mu \in \mathbb{R}^{9m}} \|\xi^{\text{full}} - P_{\text{aff}}\mu\|_{\widehat{S}}. \quad (26)$$

Moreover, if U_i^{full} and U_i^{aff} are the corresponding local velocities,

$$\sum_i \|U_i^{\text{full}} - U_i^{\text{aff}}\|_{A_i}^2 = \|\xi^{\text{full}} - \xi^{\text{aff}}\|_{\widehat{S}}^2. \quad (27)$$

The statements remain valid when $\widehat{S}$ is only semidefinite; in that case the minimizer and full solution need not be unique, but the seminorm values are well defined.

Proof. Lemma 5.1 and the full equation imply Galerkin orthogonality:

$$P_{\text{aff}}^\top \widehat{S}(\xi^{\text{full}} - \xi^{\text{aff}}) = 0.$$

For any μ , let $e = \xi^{\text{full}} - \xi^{\text{aff}}$ and $v = \xi^{\text{aff}} - P_{\text{aff}}\mu$. The cross term vanishes, so the symmetric positive semidefinite bilinear form gives

$$\|e + v\|_{\widehat{S}}^2 = \|e\|_{\widehat{S}}^2 + \|v\|_{\widehat{S}}^2 \geq \|e\|_{\widehat{S}}^2.$$

Equality is attained at $\mu = \lambda^{\text{aff}}$, proving (26). For a coefficient difference $\delta\xi$, the local load difference is $-N_i^\top \delta\xi_i$ and Lemma 2.2 gives

$$\delta\xi^\top \widehat{S} \delta\xi = \sum_i (H_i N_i^\top \delta\xi_i)^\top A_i (H_i N_i^\top \delta\xi_i),$$

which is (27); the irrelevant global minus sign drops out. $\square$

Corollary 5.3 (Affine versus classic enrichment). *Let P_{P0} denote the full-trace restriction containing only constant normal modes. Since $\text{Range}(P_{P0}) \subset \text{Range}(P_{\text{aff}})$,*

$$\|\xi^{\text{full}} - \xi^{\text{aff}}\|_{\widehat{S}} \leq \|\xi^{\text{full}} - \xi^{P0}\|_{\widehat{S}}. \quad (28)$$

No analogous monotonicity follows automatically for velocity L^2 , pressure L^2 , outlet flux, or broken H^1 errors.

6 Statements that are not established

1. **Unconditional local rank.** The materials supplied do not prove Assumption 3.2 for general Lipschitz pores or for all meshes. Numerical observation of one near-zero eigenvalue is evidence for tested cases, not a general theorem.
2. **Per-face independence implies SPD.** This implication is false as a proof principle: per-face independence does not exclude cross-face dependencies or dependencies after applying H_i . The valid global theorem is Theorem 4.2, conditional on the exact local-kernel property.
3. **Broken-pressure traction system equals monolithic FEM.** The full system in Section 6 uses independent local pressure spaces. The monolithic Taylor–Hood problem uses a globally continuous pressure space. No equivalence proof is available in the supplied material. The implemented “Exact FE-trace Schur” is a primal elimination of the monolithic system and therefore cannot by itself verify equality with the dual broken-pressure system used in Theorem 5.2.
4. **Continuum or mesh-independent convergence.** The preceding results are finite-dimensional and mesh-dependent. A continuum estimate requires stable trace-space limits and a fixed interface topology. Watershed partitions whose basin count changes with mesh size do not meet that requirement.

7 Practical algebraic checks

For each assembled pore, the following diagnostics directly test the actual hypotheses used above:

1. verify symmetry and positive definiteness of A_i ;
2. verify the numerical row rank of B_i ;
3. verify $\|L_i \mathbf{1}_{i,0n} - B_i^\top \mathbf{1}_i^p\|$ at assembly tolerance;
4. compute singular values of $H_i L_i$ and check for exactly one null direction aligned with $\mathbf{1}_{i,0n}$;
5. after global assembly, verify symmetry and positive definiteness of $\mathbf{S}$ rather than inferring it solely from geometry;
6. verify local mass, interface moment, and global boundary-flux residuals independently.

References

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